

Mohammadmahdi Gheisari

AI Engineer — LLM, RAG & Privacy-First Generative AI

Turin, Italy · Authorised to work in Italy · Open to hybrid or on-site roles
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PROFILE

AI Engineer who builds self-hosted LLM and RAG systems for environments where data cannot leave the perimeter. As CTO of an early-stage startup, I architected an on-prem RAG pipeline (Ollama, PostgreSQL/pgvector) with citation-enforced answers, shipped 5 real-time vision models to production and led a team of 6 engineers. Since January 2026 I have been building NeuroLingo, a multi-provider LLM application that keeps working fully offline. M.Sc. in Data Science and Engineering, Politecnico di Torino.

SKILLS

LLM & RAG	Anthropic, OpenAI and Gemini APIs · multi-provider routing and failover · RAG (from scratch and framework-based) · embeddings and semantic search · PostgreSQL/pgvector · Ollama · llama.cpp (GGUF)
Privacy & security	on-prem and self-hosted deployment · offline-first architecture · controls aligned with IEC 62443 / CRA
ML & vision	PyTorch · TensorFlow · OpenCV · scikit-learn · anomaly detection · one-shot and low-data learning
Engineering	Python · C++ · SQL · SQLite · Django · REST APIs · WebSockets · Docker · Kubernetes · ONNX · TensorRT · NVIDIA Jetson · GitHub Actions (CI/CD) · automated testing
Languages	English (C1) · Italian (A2) · Persian (native)

EXPERIENCE

Machine Learning Engineer — NeuroLingo (independent project) Turin · Jan 2026 – Present

- Built a provider-agnostic LLM routing layer over Anthropic, OpenAI and Gemini with cascading failover to a local GGUF model via llama.cpp, so the app runs with no network connection and no API key.
- Implemented RAG from scratch to meet mobile binary-size limits: hashing-trick embeddings (4,096-dim) in a NumPy vector store, with cosine similarity computed as a single matrix product, without PyTorch, FAISS or sentence-transformers.
- Designed the data layer (SM-2 spaced-repetition engine, repository-pattern SQLite with WAL journaling and cascading foreign keys), covered by 212 passing tests; packaged cross-platform with Flet/Flutter and PyInstaller.

Chief Technology Officer (Machine Learning & Computer Vision) — Therness S.r.l. (early-stage startup) Turin · Sep 2025 – Jan 2026

- Architected an on-prem RAG pipeline for SOP retrieval (Ollama, PostgreSQL/pgvector) with citation-enforced, grounded answers and a refusal fallback; fully self-hosted, with no customer data leaving the client's environment and controls aligned with IEC 62443 / CRA.
- Productionized 5 defect-detection models (PyTorch to ONNX/TensorRT) on GPU-edge devices, running at the sensor's full 25 FPS capture rate; benchmarked at 94 FPS unconstrained (~3.8× headroom).
- Cut inspection setup time by 97% (33 minutes to under 53 seconds) by re-engineering the workflow with Django and WebSockets.
- Led a 6-engineer team from R&D prototypes to production-grade systems and owned customer-facing technical delivery.

R&D Scientist — Computer Vision & Deep Learning — Therness S.r.l. Turin · Sep 2024 – Aug 2025

- Developed thermography-based defect-detection prototypes using one-shot and low-data learning; built data-validation, preprocessing and error-analysis workflows to benchmark models on real-world thermal video.

Data Scientist Intern — Ades Consulting & Service S.r.l. Brescia · Jul 2024 – Sep 2024

- Built a robotic weld path-planning pipeline from STL meshes, matching ~87% of reference trajectories; wrote Python tooling that shortened stakeholder review cycles.

EDUCATION

M.Sc. in Data Science and Engineering — Politecnico di Torino Turin · Mar 2026
Thesis: One-Shot Learning algorithms in computer vision tasks on thermographic acquisition

B.Sc. in Applied Statistics — University of Hormozgan Bandar Abbas, Iran · May 2017

Certifications: Machine Learning & Deep Learning Specialization (DeepLearning.AI) · Agile & Scrum (IBM)